mmWave Radar based Environment Perception:

Project with Bing Zhu

Jianan Liu
chisyliu@Hotmail.com

The Brief Project Background:

We consider using **Automotive Radar/mmWave Radar** as the main sensor for environment perception, which includes object detection and target tracking as the major purpose. The brief idea is to design deep learning based approach for object detection by using radar data, provide the detected objects. Then doing the further processing of tracking to track the objects in bird eye's view(BEV)/3D space in time. Furthermore, we could also consider designing an end to end deep learning based approach as one solution for "object detection and tracking" in BEV/3D space(input raw data or points, output track information).

The Aimed Publication:

We consider to submit and get accepted on the following conferences and journals:
IEEE ICRA, IEEE IROS, FUSION, IEEE IV, IEEE ITSC, IEEE RadarConf
IEEE Transactions on Intelligent Transportation Systems, IEEE Sensor Journal, IEEE Transactions on Intelligent Vehicles, etc.

Scenario, Dataset, and Sensor Set:

Based on combination of different conditions/prerequisites/configurations, we could come up with different solutions for perception.
We basically could have:

1. Indoor v.s. Outdoor;
2. More confined area based scenarios v.s. Highway and city traffic scenarios;
3. Radar only v.s. Radar + Camera;
4. Detection points as the data available from radar v.s. Raw data(e.g. IQ data, or processed range-Doppler map, range-azimuth heat map).

It is always better if we could access into raw level data from radar, since there are lots of information can be extracted from raw data. However, additional statistical signal processing, e.g. FFT, CFAR, MUSIC, ESPRIT, etc, need to be provided for estimation of range bin, Doppler bin, and angle bin in an accurate enough level. On the other hand, the detection points is more straight forward to use but lots of hidden features may already been filtered out. We can start with detection points data which can be available from open source dataset using conti radar(e.g. nuScenes), to let student get familiar with PyTorch, object detection, radar data, etc. Later we can consider moving towards real measured raw level/processed data from high resolution 4D radar, e.g. Navtech, or high resolution 3D, e.g. TI radar, with focus on outdoor environment perception(e.g. RADIATE, CARRADA, CRUW).
We can start with radar only object detection and target tracking, then consider adding more sensor modality, e.g. mono-camera, IR camera(infrared camera), etc, as complementary.

<u>Possible Ideas and Tasks to Do</u>:

1. <u>Object Detection</u>

We consider designing the deep learning based object detector using radar date only, output the estimated 2D bounding box on the bird eye's view(BEV) with the pose information, i.e. x, y, and orientation, and shape information, i.e. length and width.

1.1. The basic idea is use one stage object detector(Anchor Based, Anchor Free), RetinaNet/YOLOv3,v4/FCOSv1,v2/CenterNet_v1(Center Points as Object), etc, to load the "image like radar data"(either detection points or range-azimuth heat map which contains between possible semantic like information and pose information of the multiple objects) and do the object detection. However, radar has lots of the problems like sparsity, instability due to multi-path, lack of semantic information, etc, which makes it possibly hard to be applied on one stage object detector which was originally designed for image based object detection task directly.

1.2. We might also need to consider how to use attention mechanism in space, channel, etc, especially consider how to combine the distance information which can be provided perfectly even when the object is far away of radar, with attention mechanism to make the network suitable for detecting the distant objects.

We could consider using some ordinary commonly used attention mechanisms: channel attention, pixel-wise attention, self-attention, etc, and even consider some particular attention(assigning weights mechanisms) for radar based object detection, e.g.

a). 一种可能的思路是：对于 BEV 上的信息，设计一种产生 spatial-wise attention 的方案来对远距离的目标所提取的 feature 给予更大的 weight（比如某种和 measured distance 有关的 function，距离远的返回的值较大，从而可以将该 function 的返回值作为 spatial-wise attention 或者 conv kernel-wise attention 的 weight）。

b). 第二种可能的思路是：（如果需要投影回 front view 的情况）对于投影回 front view image 上的信息，设计一种产生 spatial-wise attention 的方案来对处于越图像中央的部分的目标所提取的 feature 给予更大的 weight（比如对整张图使用比较扁平的 3D Gaussian distribution，越靠近 front view image 中间的区域将根据 Gaussian distribution 的函数生成更大的 weight，考虑使用 Gaussian distribution function 的原因是 Gaussian distribution 更平滑）。

c). 第三种可能的思路是：为了处理远近不同的目标带来的更好的 extract multi-scale features for multi-scale objects 问题，通常我们比较传统的思路处理这种 multi-scale feature extraction 的方案就是比如使用 FPN。我们可以使用 FPN+Attention 的思路（即 ASFF 的思路），即对于 FPN 结构中产生不同 scale 的 feature 的 feature map 给予不同的 weight 再 fuse 到一起。

d). 第四种可能的思路是：还是为了处理远近不同的目标带来的更好的 extract multi-scale features for multi-scale objects 问题，另一种我们比较传统的思路处理这种 multi-scale feature extraction 的方案就是在多个并联的 path 上用不同 kernel size 的 dilated conv(空洞卷积或者叫做扩张卷积，为了可以用相同数量的参数得到更大的感受野)然后再 fuse 到一起，比如 RFB-ESRGAN。那么自然的，我们也想到可以对多个不同 kernel size 的 dilated conv 直接应用 conv kernel-wise attention（见 SKNet），对于不同 size 的 conv kernel（或者不同 dilated rate 的 dilated conv kernel）并联再 fuse 的结构，每一个不同 size 的 conv kernel 分配不同的 weight 造成对于不同的感受野（不同的 size 的 feature）有更好的抓取效果。

Besides, we could also check this paper: 2020.Attention Models for Point Clouds in Deep Learning: A Survey (https://arxiv.org/abs/2102.10788) to see how the attention have been used in Lidar point cloud and think about what could be re-used in radar data.

1.3. We may also want to consider better way to modify the network architecture or to use dilated convolutional filter for bringing better capability to extract the features in multi-scales, for better supporting the detection of distant objects.

1.4. Adding the **mechanism to modeling the uncertainty(variance) of bounding box and estimate the uncertainty information** in accurate way, thus supporting better performance in the tracking stage. Firstly need to read this paper: 2020.A Review and Comparative Study on Probabilistic Object Detection in Autonomous Driving, getting better understand.

建模 uncertainty 已知的方案包括：

a). 2019.Gaussian YOLOv3: An Accurate and Fast Object Detector Using Localization Uncertainty for Autonomous Driving 这篇论文使用的方案，利用最大化 ground truth 在预测的均值和方差描述的高斯分布下的 pdf 函数的连乘的方式来 maximize likelihood 来建模 bounding box 参数为 Gaussian 分布的均值和方差。

b). 2020.Probabilistic Oriented Object Detection in Automotive Radar, 2020.Uncertainty Estimation in One-Stage Object Detection, 2017.What uncertainties do we need in bayesian deep learning for computer vision, 和 2019.Bounding Box Regression with Uncertainty for Accurate Object Detection 这几篇论文都使用的相同的方案，利用“求确定的 ground truth 真值对应的 Dirac distribution 和预测的 Gaussian distribution 之间的 KL 散度”来建模 bounding box 参数为 Gaussian 分布的均值和方差。

c). 我们自己想到的新 idea，用 Wasserstein 距离替代 KL 散度去定义一个“高斯分布（或者任意分布）的随机变量预测框和确定值的 ground truth 框（Dirac 分布）之间的 Wasserstein 距离”，把这个“Wasserstein 距离”作为新的带有 uncertainty 信息的 localization loss 项，来建模 bounding box 参数为 Gaussian 分布（或者任意分布）的均值和方差。Wasserstein 距离和 KL 散度区别见：

https://zxth93.github.io/2017/09/27/KL%E6%95%A3%E5%BA%A6JS%E6%95%A3%E5%BA%A6Wasserstein%E8%B7%9D%E7%A6%BB/index.html

https://zhuanlan.zhihu.com/p/74075915

d). 2020.Generalized Focal Loss: Learning Qualified and Distributed Bounding Boxes for Dense Object Detection 中的方案，建模 bounding box 参数为任意分布。

那如何利用预测出来的 uncertainty 信息呢？已知的方案包括：

a). 2019.Gaussian YOLOv3: An Accurate and Fast Object Detector Using Localization Uncertainty for Autonomous Driving 这篇论文使用的方案,在 NMS 的过程中通过 Cr 因子的大小对预测框进行筛选从而利用这个新引入的 bbox 的不确定性(方差)信息。

b). 我们的新 idea：可以考虑从 IOU 的角度出发，定义基于高斯分布的 IOU（i.e. Gaussian IOU）从而利用不确定信息，进一步得到相应的定位损失函数（i.e. Gaussian IOU Loss）取代原来的损失函数进行训练。

c). 我们的新 idea：将预测出的每一个位置和长宽已经建模成高斯分布的 bbox 作为 tracking 的输入，由于不通过 NMS 去筛选，所以一个目标可能会含有多个 bbox 作为 detections，然后在数据关联的过程中去选择最可能的 detection 去对目标进行 update。这样做的好处是，每一个预测出的 bbox 已经含有 object detector 预测出的 bbox 方差信息了，不需要人为假设方差信息对目标进行 update。

1.5. Absorb the latest **anchor free object detector** designing into our radar based object detector.

1.6. Knowledge distillation for radar based object detection. Design a new or reuse existing radar + camera fusion based object detector network, employ it as teacher network. Treat our

proposed radar based object detector network as student network, apply knowledge distillation to train our proposed radar based object detector network.

1.7. Apply **LSTM together with CNN for radar based object detection**, e.g. Use CNN to extract features for range-azimuth heat map on very frame, then feed multiple feature maps on several consecutive frames(could be converted into histogram, maybe?) into LSTM to extract Doppler variation along time axis to estimate the objects for each frame. 注意这里的 several consecutive frames 可以是当前检测帧之前的连续 N 帧（N 的值理论上需要足够长保证比如 **Micro Doppler** 信息的 pattern 可以被显现出来）。CRUW 数据集中给出的每个 batch 的数据的 snippet(一小段时间)长度为 16(即 16 帧)，这个时间长度肯定是足够 Micro Doppler 被显现出来了。

LSTM 的输入是 t-n 到 t 时刻的 Cartesian Coordinate 的 BEV feature map，输出是 t-n 到 t 时刻的 Cartesian Coordinate 的 BEV feature map。

LTSM 的结构解析见：

https://colah.github.io/posts/2015-08-Understanding-LSTMs/

1.8. Apply **transformer** based deep learning approach for radar based object detection, e.g.

a). one possible approach is, use CNN to extract features for range-azimuth heat map on single frame, then convert into vector/sequence as how the embedding is done in Transformer. Then feed into "**DETR like**" structure to output result of classification and bounding box regression.

可以参见 facebook 设计的第一个把 Transformer 和 2D Object Detection 相结合的论文 DETR 是如何设计的（DETR 直接在 2D Image 上抓取 feature map 之后拉平，结合 2D (x, y)俩个方向上的位置编码信息都赋给 Query 与 Key 之后，直接用 google Transformer 原论文的 Multi-self attention+feed forward network（其实就是极其简单的多层感知机, MLP 结构）构成的 encoder 结构和 decoder 结构连接 detection head 出 classification 和 2D bounding box regression 的结果。另外还要注意 DETR 是 anchor box free 的检测器，为了可以让 detection head 预测出的 classification 的结果与 regression of 2D bounding box 的结果一一对应上，论文定义"从真值索引(真实的 class 和 bounding box)到预测值索引(预测的 class 和 bounding box)的一种度量距离(e.g. 分类结果的差异+GIOU)"在所有可能的 classification 的结果和 regression of 2D bounding box 的结果构成的组合的 sum 作为 cost function，这样就可以像 tracking 中做 data association 那样用 auction algorithm/Hungarian algorithm 实现 GNN 的方式找到最小的 cost 值对应的那个组合，来求取最可能的 classification 的结果与 regression of 2D bounding box 的结果的组合。）。

关于 DETR Object Detector 详见论文：

2020.End-to-End Object Detection with Transformers

相关分析（包括什么是 multi self-attention，如何用 multi self-attention 构成 Transformer 结构，如何用 Transformer 结构直接对单帧图像操作实现 2D object detection，segmentation 等任务）：

https://mp.weixin.qq.com/s/HU1wXzPH6yZEbMPstES1CA

b). another possible approach is, use CNN to extract features for range-azimuth heat map on every frame, then feed multiple feature maps on several consecutive frames(could be converted into histogram, maybe? Or could flatten the feature maps on several consecutive frames into vector then leave it into transformer to exact correlation/dependence relationship along all elements) into transformer to extract Doppler variation along time axis and correlation/dependence relationship along all elements, then estimate the objects for each frame. 注意这里的 several consecutive frames 可以是当前检测帧之前的连续 N 帧（N 的值理论上需要足够长保证比如 **Micro Doppler** 信息的 pattern 可以被显现出来）。然后可以再将

Transformer 抓取出来 along time axis 的 feature maps 再 fuse 回当前帧用 CNN 抓取出来的 feature maps 中，然后再把 fused feature map 上采样，再送到 detection head 中。这种网络结构可以参照 TransUNet 的结构：

Paper: 2021.TransUNet: Transformers Make Strong Encoders for Medical Image Segmentation

https://arxiv.org/abs/2102.04306

https://zhuanlan.zhihu.com/p/350271375

又或者按照论文 2021.Channel Boosting Feature Ensemble for Radar-based Object Detection 中的结构**直接用多个 encoder(feature extraction) branches 抓取不同帧的 feature maps 然后直接送到 DETR Transformer 中实现 Object Detection。**

值得注意的是，方案 a)中是利用 DETR 直接抓取单帧 2D image field 的 feature map 然后拉平成为一个 vector(sequence)进入 Transformer 处理，即 **Feed the Feature Maps Extracted using Existing Solution(e.g. Range-Azimuth Heatmap 进入 Encoder Decoder One Stage Object Detector 的 Encoder 特征抓取器去抓取 feature map) Directly into Transformer(DETR)**。

而方案 b)中我们所考虑实现“抓取时间序列上的 feature 形成 vector(sequence)进入 Transformer 处理得到新的 feature map 再和 CNN 抓取的 2D image field 上的 feature map 进行 fuse。再把 fused feature map 上采样，再送到 detection head 中”，或者“抓取时间序列上的 feature 形成 vector(sequence)进入 Transformer(DETR)处理直接生成 object detection 的 bounding box”。这俩种思路有本质上的区别。

c). One more possible approach is, **Extract Feature Maps from Single Frame Radar Data using Multiple Existing Solutions(e.g. 首先 Range-Azimuth Heatmap 进入 Encoder Decoder One Stage Object Detector 的 Encoder 特征抓取器去抓取 feature map。 然后 Radar Detection Points 进入 PointNet++抓取每个 Point 的特征，再把所有 Points 的特征拼接到一起形成另一个 feature map 。) then Feed multiple Feature Maps from different solutions into Transformer(DETR)。**

d). One more possible approach is, **Extract Feature Maps from Single Frame Radar Data using Multiple-Field Representations of Radar Data(e.g. 论文 2021.Channel Boosting Feature Ensemble for Radar-based Object Detection 中使用 Range-Azimuth Heatmap 的三种不同图像对比度的 field represenations: RGB 形式，LUV 形式，LAB 形式。我们比如可以考虑对 Range-Azimuth Heatmap 做 Wavelet Transformation 提取高频信息，然后把高频信息作为另一种 field representations。) then Feed Feature Maps from different field representations into Transformer(DETR)。**

注意：上述的结合 Transformer 的方案中，可能都要改俩个地方：

1) ．一个是给 Transformer 进数据的时候图像或者 feature map 要先经过（Image-to-tokens 模块）先缩小再分 patch。

见 CeiT 论文 2021.Incorporating Convolution Designs into Visual Transformers 和这个文章：

https://zhuanlan.zhihu.com/p/361112935

以此来缩小网络大小和要耗费的 GPU 运算量。

2). 另一个就是用 CPVT 这篇文章提出的这个模块来自动生成一个位置编码加在原输入 patch 序列后再给 Transformer，这种方案利用输入序列本身自适应的生成针对输入序列的位置编码，不需要针对不同大小的输入序列每一次都重新设计位置编码。

见

https://mp.weixin.qq.com/s/T5rq--w3WDJk26L72sMBOQ

和

https://zhuanlan.zhihu.com/p/353229685

CPVT 代码见这篇博客：https://mp.weixin.qq.com/s/3wqnVrxfd0Smw8Mexp7Ptg

1.9. **Apply wavelet transformation**, e.g. 有没有可能对 range azimuth heatmap 做 wavelet transformation 得到各个频率的信息，之后再改变 detector 的 backbone 和 neck 的结构来更合适的利用这些信息做 Object Detection? （为了更好的利用小波变换之后的信息，我们可以试着对比如 range azimuth heatmap 或者其他类型的"image like"的用作网络输入数据的雷达数据先进行 wavelet transformation，观察小波域的高低频分量的样子，看看是否变换后得到了一些更加明显的特征：例如是否一些 heatmap 中 object 的边缘位置在小波域的高频分量中更加明显了）比如直接将 Wavelet Transformation 之后的高频分量提取特征之后和普通 range azimuth heatmap 提取的特征直接 concatenate 到一块再做 object detection?

1.10. **数据原来没有的其他相关特征信息 fuse 到数据上再做 object detection**。注意这样的思路是很常见的。PointPainting（PointPainting 是基于 C+L 的网络本质上就是先用一个训练好的做 Semantic Segmentation 网络得到 Camera 的 Segmentation 的结果，也就是 Camera Image 上的每一个 pixel 都给出语义信息，然后把 Lidar Point Cloud 的每一个 point 都投影到 Image 上获得对应每一个 point 的语义信息，然后将这个语义信息当做一个新的维度 concatenate 到 point 上。之后就可以用任意的基于 Lidar 的 Object Detector 做 Object Detection 了。唯一的区别在原来的 Lidar point cloud 上多了一维的信息。）和 Pseudo Lidar（基于双目图像，估计深度信息再把深度信息加载到图像上做成"伪点云"做 object detection）都是这种逻辑。

a). 像 PointPillar 的那种"不同方案联合使用的"类似思路，我们可以考虑用 Points 和 Points 离散化之后得到的 BEV pseudo image 各自提取 features，设计一个 two stage object detector: first stage RPN 网络先把 Points 离散化之后得到的 BEV pseudo image（或者 range azimuth heatmap）用 Encoder Decoder 结构的 Encoder 提取特征，进行训练令其可以生成 ROI proposals。之后 second stage 把每个 BEV pseudo image（或者 range azimuth heatmap）估计出来的 ROI proposals(Bounding box)对应区域的 Points 用 PointNet++提取 Feature 再 fuse(concatenate)进来再 refinement bounding box 和分类。

注：这个方案实现的 bottleneck 在于数据集。nuScenes 数据集提供 radar detection points 和 3D bounding box 的 ground truth，但是 radar detection points 过于 sparse，很可能效果很差。RadarScenes 数据集提供相对比较密集的 radar detection points，但 ground truth 却不是 bounding box 而是 points 的 instance segmentation。对于 NuScenes 数据集的问题解决方案可以是多帧积累线性插值得到更密集的 radar detection points。对于 RadarScenes 数据集的问题解决方案可以是用某种算法先标注 Ground truth bounding box。

b). 又或者使用像 PointPainting 那样把语义分割的信息加载在 radar detection point 上再 fuse 给 range-azimuth heatmap 或者"detection points 离散化产生的 BEV pseudo image"做 Object detection 或者 instance segmentation（比如

b).1. First stage object detector: 用 PointNet++给 radar detection points 提取语义产生一张 feature map，同时用 CNN 对 range-azimuth heatmap 或者 BEV pseudo image 提取另一张 feature，之后再把俩个 feature map 来 concatenate 到一起做基于带有隐含的语义特征的 range-azimuth heatmap 或者基于带有隐含的语义特征的 BEV pseudo image 的 object detection 或者 instance segmentation。

b).2. Second stage object detector: first stage 先用 PointNet++给 radar detection points 提取每个 point 对应的语义信息）。对该 first stage 网络进行训练，之后可以产生每个 detection point 对应的语义信息。然后再把每个 Detection point 的语义信息 concatenate 到 Radar 的 range-azimuth heatmap 或者 BEV pseudo image 上，再训练 second stage 网络对带有显示语义信息的 Radar 的 range-azimuth heatmap 或者 BEV pseudo image 做 object detection 或者 instance segmentation。这个思路其实和基于 Lidar Multi-View 的 object detector，MVLidarNet，

比较相像。
注：这个方案可以用 RadarScenes 数据集实现。RadarScenes 数据集提供相对比较密集的 radar detection points，ground truth 是 points 的 instance segmentation。
1.11. C+R based object detection 方案：
1). C+R，基于 PointPainting 的方案。把 Radar Point Cloud 投影回 Camera Front View，找到每个 Point 对应的 Pixel 然后把相应的语义信息 fuse 到 Point Cloud 上，再做基于 PointNet 的 Object Detection(具体方案详见 PointPainting 论文和 Pointillism 论文)。
2). C+R，基于 PointPainting 的方案。把 Radar Point Cloud 投影回 Camera Front View，找到每个 Point 对应的 Pixel，再把 Radar Point Cloud 离散化之后变成 BEV Pseudo Image，然后把相应的语义信息 fuse 到 BEV Pseudo Image 上，再做基于 CNN(e.g. YOLOv4, CenterNet)的 Object Detection。
3). C+R，基于 PointPainting 的方案。把 Radar Point Cloud 投影回 Camera Front View，找到每个 Point 对应的 Pixel，再把 Radar Point Cloud 进行 clustering 之后再用基于 PointNet 的 Classifier+Regressor 来 estimate 每个 clustering 里面 Points 对应的 bounding box 和 class label。
4). C+R，基于 Frustum PointNet 的方案。做一个 two stage object detector: 先训练 first stage 网络，把 camera image 先用某个网络做 2D object detection，然后把 2D bounding box 投影到 3D space 形成 frustum，再把 frustum 投影到 radar 的 BEV 上。这时再训练 second stage 网络，对每个 frustum 内的 radar detection points 进行 clustering 聚类，然后把每一个 clustering 之后的结构送到基于 PointNet++的网络内进行 classification 与 bounding box regression。
为什么这样做可能有好的效果？因为 camera 测距不准但测角度准且 camera 的抖动多数在 elevation 上而不是 azimuth 上，所以 camera 投影形成的 frustum 内的 points 和 frustum 外的 points 理论上应该很大机会不属于同一个 object。而 radar 测距准而测角不准，所以直接 clustering 会有角度上相近的 points 不太容易区分是否属于不同 objects，而现在 frustum 已经在角度上区分开 objects 了，所以 radar 的 clustering 只用在 range 上分开 objects 即可，这是 radar 擅长的。
5). C+R, 基于 4).这种方案扩展得新方案：做一个 two stage object detector: 先训练 first stage 网络，camera image 也用某个网络做 3D object detection，然后把 3D object detection 的结果的对应的 2D bounding box(即框住 3D bounding box 最外侧的一个 2D bounding box)投影到 3D space 形成 frustum，再把 frustum 投影到 radar 的 BEV 上。这时再训练 second stage 网络，这时再对每个 frustum 内的 radar detection points 进行 clustering 聚类，然后把每一个 clustering 之后的结构送到基于 PointNet++的网络内进行 classification 与 bounding box regression。将该基于 PointNet++的网络生成的 BEV bounding box 和 camera image 生成的 3D bounding box 投影到 radar 的 BEV 上的 2D BEV bounding box 进行基于 detection level 的 fuse(e.g. 用某种 NMS 或者相似的方案)得到 final 2D BEV bounding box output。
6). C+R, 基于论文 2020.Low-level Sensor Fusion for 3D Vehicle Detection using Radar Range-Azimuth Heatmap and Monocular Image 和 2020.GRIF Net: Gated Region of Interest Fusion Network for Robust 3D Object Detection from Radar Point Cloud and Monocular Image 这俩篇论文提到的 **cross view fusion**(也就是类似于 2020,2021 年以来 Lidar point cloud based object detection 特别流行的 multi-view fusion 的方案)方案来设计的新方案。我们考虑用 camera front view 的 image 和 radar BEV 的 detection points 作为输入数据（可用 nuScenes data），设计一个 one stage object detector。用一个 CNN backbone 和 PointNet++分别提取 front view image 和 BEV detection points 的 features，然后直接 cross view fuse 或者通过某种 attention 的方案结合 concatenate 实现 cross view fuse 到一起生成一个 feature map，然后在这个新的 feature map 上根据预先设置的 3D anchor boxes（anchor boxes 设置可以利用

detection points 作为 anchor point 来设置，即仅在有 detection points 的位置设置多个 anchor boxes）直接回归最后的 3D bounding box。整体流程可参考提到的俩篇论文的网络结构的 second stage 部分（俩篇论文的网络结构原论文的 figure 5 与 figure 2）

7). C+R based object detection 与 transformer 结合的方案还没怎么被研究过，可以考虑把前述纯基于 radar 数据的 object detection 与 transformer 结合的方案移植来 C+R based object detection. 可能的方案比如把之前 1.8.b)., 1.8.c)., 1.8.d).提到的将 Transformer 结合“多帧信息”，“多种方案抓取的 feature maps”，“同一帧信息的多种表征(e.g. Wavelet frequent domain 和 time domain)”的方案变为：将 Transformer 用在结合 multi-view feature maps 上，换言之即用 Transformer 替代 1.11.6).中提出的“直接 concatenation 或者某种 attention 的方式”把 C 在 front view 的 feature map 和 R 在 BEV 的 feature maps 给 cross view fusion。

8). C+R based object detection 结合 uncertainty modeling 的方案还没怎么被研究过，可以考虑把前述纯基于 radar 数据的 object detection 与 transformer 结合的方案移植来 C+R based object detection.

以上 8 种方案可以与前面的 1.10.a). 1.10.b).俩种基于纯 radar 数据的方案进行比较。

9). C+R object detection + target tracking，依然基于 Frustum PointNet 的方案：做一个 two stage object detector: 先训练 first stage 网络，还是把 camera image 先用某个网络做 2D object detection，然后把 2D bounding box 投影到 3D space 形成 frustum，再把 frustum 投影到 radar 的 BEV 上，frustum 内的 points 和 frustum 外的 points 在 azimuth 上面就被分开了。接下来(**使用某种方案，具体是什么方案生成多个 patches？** 比如可以是对每个 frustum 内的 radar detection points 进行 clustering 聚类)在每个 frustum 内生成多个 patches(允许不同 patch 间有重叠)，这时再训练 second stage 网络把每个 patch 都用基于 PointNet++的方案去生成 2D bounding box with uncertainty information。生成的 2D bounding box with uncertainty information 给后续的 tracking 提供多个 measurements with accurate variance information，从而可以直接用 Point Target Tracking 的 Bayesian filter(对于 Bayesian filter，data association 可以用 GNN 或者 JPDA)或者 RFS filter 的 framework 进行 update。所以这种思路实际是在之前提出的 1.11.4).或者 1.11.5)基础上对所有产生的 bounding box 同时再估计 uncertainty，之后再将这些 variance 信息输入后续 Bayesian tracking filter 中来更好的实现 tracking。

2. <u>Target Tracking</u>

2.1. Apply the traditional Bayesian filter(e.g. Kalman, EKF, UKF/Sigma Point Filter) on the single linkage clustered detection points directly, together with GNN as data association approach.

2.2. Apply the traditional Bayesian filter(e.g. Kalman, EKF, UKF/Sigma Point Filter) on estimated bounding boxes on BEV, together with GNN as data association approach.

2.3. Apply the traditional Bayesian filter(e.g. Kalman, EKF, UKF/Sigma Point Filter) on estimated bounding boxes on BEV with better/accurate uncertainty(covariance) description, together with GNN/JPDA as data association approach.

2.4. Apply RFS filter like PHD filter, LMB, PMBM filter to replace Bayesian filter(in all the solutions above). 几个可选方案：

a). **trajectory PHD/CPHD(TPHD/TCPHD)**

b). **trajectory PHD/CPHD which provides ID continuity together with IMM motion model**(Jump

Markovian System), e.g. See paper 2020.The Trajectory PHD Filter for Jump Markov System Models and Its Gaussian Mixture Implementation.

c). LMB

d). **trajectory PMBM(TPMBM)**

e). **trajectory PMB(TPMB)**. See paper: 2020.Trajectory Poisson multi-Bernoulli filters

理论上 TPMB 最快最准，TPMBM 次之。并且据这篇论文(paper: 2021.Multi-Object Tracking using Poisson Multi-Bernoulli Mixture Filtering for Autonomous Vehicles 称 PMBM 的 point target tracking 方案在 Waymo dataset 和 argoverse dataset 上性能远超 Bayesian filter based 和 deep learning based tracking 方案，并且可以有效的以 20Hz 的频率运行在笔记本电脑上。

注意：All the PHD, TPHD, PMBM, TPMBM point target Matlab code could be available here: https://github.com/Agarciafernandez/MTT

We need to rewrite using Python.

对于上述 idea，可以映射到以下具象的具体可做的方案：

A). Object Detection 扩展到 Tracking 的直接方案一（基于 detection points 的 measurement，point target model）：对于当前使用的 CRUW 数据集，因为其 ground truth 表示为点，所以天然的非常适合 point target tracking 的方案，一个可以做的 idea 如下：

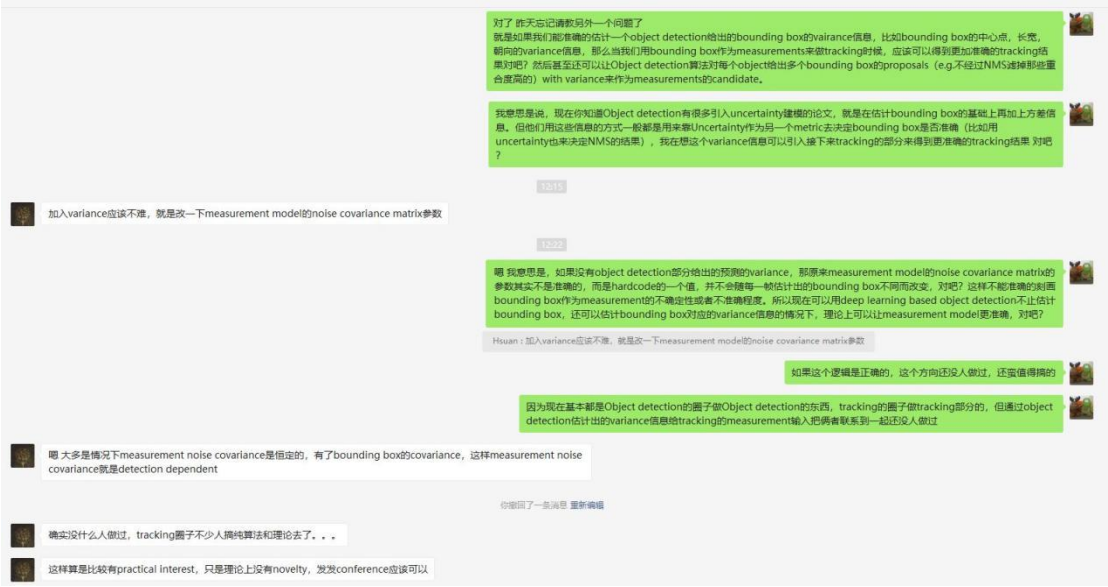
用之前设计并训练好的 BEV object detector 估计 targets 的点位置（作为高斯分布的均值）和相应的估计出的高斯分布的方差（即估计出的 confidence map 的方差）作为 detections。这样形式的 detection 结果非常适合直接作为点目标的 detection 来做 tracking。但是要注意，因为 CRUW 数据集的 ground truth 仅有 target 的点位置而没有 ID 信息，所以我们这里做的 tracking 其实只需要对 detection 提供的 target 的 location 信息做 refinement（因为只能测试 target 的 location 信息准确与否，而不能测试 ID continuity 是否准确的）。所以一个简单的思路为，直接用 PHD 或者 CPHD filter（这种不提供 ID continuity 的 RFS filter）给 detections 做 filtering 即可得到对 target 的 location 信息的 refinement(update)，从而得到更准确的 target localization estimation。同时因为估计出的 detection 结果带有语义信息（class label），所以可以对不同的 class 的 target 使用不同的 motion model，例如对于 car 的 motion model 可以考虑用 Constant Velocity 或者 Constant Turning Rate，pedestrian 的 motion model 可以考虑用 Random walking，cyclist 的 motion model 可以考虑用 IMM。

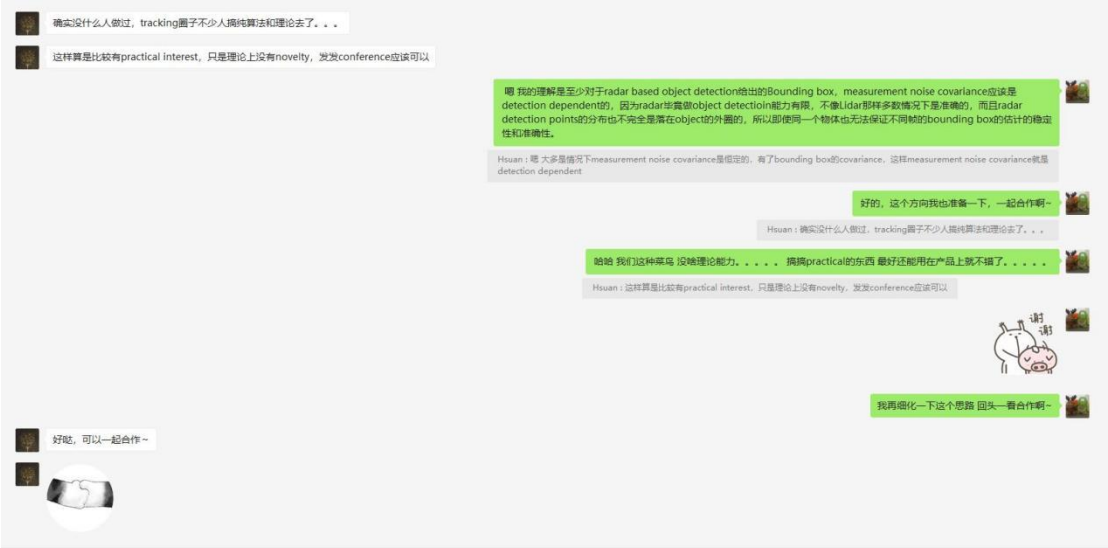
B). Object Detection 扩展到 Tracking 的直接方案二（基于 detection points 的 measurement，extended target model）：对于当前 RadarScenes 数据集，因为其 ground truth 表示为来自每一个 object 的多个 detection points 的 instance segmentation 信息，所以天然的非常适合 extended target tracking 的方案，一个可以做的 idea 如下：

首先通过某种方案(比如上述 1.10.b).)估计出每个 Point 的语义信息。之后再做 RFS filter based extended target tracking，这样这些 detection points 就已经包含语义信息了。比如用 Bayesian filter, PHD 或者 PMBM 做 extended target tracking。 对于 car 的 points 的 measurements model 可以考虑用 rounded rectangular(四角为圆形的长方形)，对于 pedestrian 的 points 的 measurements model 可以考虑用 elliptic(椭圆)。对于 car 的 motion model 可以考虑用 Constant Velocity 或者 Constant Turning Rate，pedestrian 的 motion model 可以考虑用 Random walking，cyclist 的 motion model 可以考虑用 IMM。另外注意因为已经做了 instance segmentation 了，所以相当于 clustering 已经做好了，只需要用每个 instance 包含的所有 points 用对应的 measurement model 进行 update 就可以了。

C). Object Detection 扩展到 Tracking 的直接方案三(基于 bounding box 的 measurement，point target model)：吸收 uncertainty 的 bounding box 作为 measurement 来做 tracking 这个 idea，

具体一个可以做的 idea 如下：





2.5. Apply deep learning approach, e.g. LSTM, **Informer**(Best paper in AAAI21. See paper: 2021.Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. Code: https://github.com/zhouhaoyi/Informer2020) to replace filter and/or data associator, for radar based tracking in BEV.

2.6. Apply **transformer** based deep learning approach, to replace filter and/or data associator, for radar based tracking in BEV.

3. *Perception = Object Detection + Target Tracking 整体构架方案的选择和比较的 idea*

总的来看，组合以上 *Object detection* 跟 *Target Tracking* 部分提出的具体研究点，得到我们的主要框架以实现以下几种 radar only 或者 R+C 的 perception 方案构架为主，并进行比较：

a). R/R+C deep learning based object detection + Bayesian filter tracking

b). R/R+C deep learning based object detection + trajectory PHD/CPHD filter tracking

c). R/R+C deep learning based object detection + trajectory PMBM/PMB filter tracking

d). R/**R+C deep learning based object detection and data association "appearance features" + tracking**

e). R/R+C deep learning based object detection + trajectory PHD/CPHD filter tracking + deep learning based data association

f). R/R+C deep learning based object detection and tracking(ID estimation)**(用深度学习模型直接输出已经认为足够准确的 object detection bounding box + ID)**

g). **R/R+C deep learning based object detection + trajectory PMBM/PMB filter tracking + deep learning based appearance features extraction for data association （至今还没有人做过，值得做）**

用 python 实现上述方案之后比较效果

项目实施计划：

Tasks until 20210512

a). Radar Data and Intro to Radar Signal Processing and Data Processing for ADAS/AD

CARRADA Dataset:

论文：https://arxiv.org/abs/2005.01456

相关的 SDK tool 代码：https://github.com/valeoai/carrada_dataset

数据下载：https://arthurouaknine.github.io/codeanddata/carrada

RADIATE DATASET:

论文：https://arxiv.org/abs/2010.09076

相关的 SDK tool 代码：https://github.com/marcelsheeny/radiate_sdk

详细信息：http://pro.hw.ac.uk/radiate/

下载：http://pro.hw.ac.uk/radiate/downloads/

CRUW Dataset:

Baseline paper：

https://openaccess.thecvf.com/content/WACV2021/html/Wang_RODNet_Radar_Object_Detection_Using_Cross-Modal_Supervision_WACV_2021_paper.html

相关的 SDK tool 代码：https://github.com/yizhou-wang/cruw-devkit

Baseline paper code：https://github.com/yizhou-wang/RODNet

详细信息：https://www.cruwdataset.org/introduction

下载：https://www.cruwdataset.org/download

NuScenes Dataset:

下载：https://www.nuscenes.org/

Paper: https://arxiv.org/abs/1903.11027

Development Kit: https://github.com/nutonomy/nuscenes-devkit

RadarScenes Dataset:

下载：https://radar-scenes.com/

Paper: https://arxiv.org/abs/2104.02493

Development Kit: https://github.com/oleschum/radar_scenes

RADDet Dataset:

下载：https://drive.google.com/drive/folders/1v-AF873jP8p6waChF3pSSqz6HXOOZgkC

Paper：https://arxiv.org/abs/2105.00363

Development Kit：https://github.com/ZhangAoCanada/RADDet

下面这篇论文详细解释了 Radar Signal Processing Pipeline 产生数据的过程，Radar 数据的形式，Radar 与 Camera 的 Calibration 等问题：

2021.RaDICaL: A Synchronized FMCW Radar, Depth, IMU and RGB Camera Data Dataset with Low-Level FMCW Radar Signals

还有一个 Presentation 详细解答了 automotive radar signal processing 和其在 ADAS/AD 系统中的相关内容：

2020.Automotive Radar – A Signal Processing Perspective on Current Technology and Future Systems
https://cloud.gardill.net/s/tjoSLSB7fXWTEBb

了解之后主要试图回答问题：
1. 这些数据集有什么区别？每一个数据集的形式为何？Ground truth label 又是什么形式？
2. 分别可以用来做什么任务？
3. 以你的理解，我们可以选用哪个数据集？
4. 什么是 Camera 和 Radar 间的 Calibration？如何使用 Calibration 的结果？

b). Generic (2D image based) object detection using deep learning
阅读以下论文:
2019.A Survey of Deep Learning-based Object Detection
2019.Deep Learning for Generic Object Detection: A Survey
2019.Object Detection in 20 Years: A Survey
to get basic understanding of generic (2D image based) object detection.
并重点理解以下几个模型 Faster R-CNN, YOLOv3/v4, FCOS, CenterNet. 思考以下这些内容：
1. 这几个模型都是做什么的，区别在哪里
2. 这些模型中各自最重要的部分是什么

c). Radar based Object Detection using Deep Learning
阅读以下论文
2020.Vehicle Detection With Automotive Radar Using Deep Learning on Range-Azimuth-Doppler Tensors
https://openaccess.thecvf.com/content_ICCVW_2019/papers/CVRSUAD/Major_Vehicle_Detection_With_Automotive_Radar_Using_Deep_Learning_on_Range-Azimuth-Doppler_ICCVW_2019_paper.pdf
2020.CNN based Road User Detection using the 3D Radar Cube
https://arxiv.org/abs/2004.12165
2020.Probabilistic Orientated Object Detection in Automotive Radar
https://arxiv.org/abs/2004.05310
2019.Experiments with mmWave Automotive Radar Test-bed
https://arxiv.org/abs/1912.12566
2020.RODNet: Radar Object Detection using Cross-Modal Supervision
https://arxiv.org/abs/2003.01816
https://arxiv.org/abs/2102.05150
代码：https://github.com/yizhou-wang/RODNet
2020.RAMP-CNN: A Novel Neural Network for Enhanced Automotive Radar Object Recognition
https://arxiv.org/abs/2011.08981
2020.Radar-based 2D Car Detection Using Deep Neural Networks
https://ieeexplore.ieee.org/document/9294546
2021.RADDet: Range-Azimuth-Doppler based Radar Object Detection for Dynamic Road Users

https://arxiv.org/abs/2105.00363

回答下面这些内容：
1. 这几篇论文都是做什么的
2. 区别在哪里
3. 这些论文提出的主要的方案是什么
4. 你有什么理解不了的问题
5. 如果让你实现论文的代码，你觉得还有哪些具体实际的问题不理解需要帮助

d). Lidar based Object Detection
2020.3D Point Cloud Processing and Learning for Autonomous Driving
https://arxiv.org/abs/2003.00601
2019.PointPillars-Fast Encoders for Object Detection from Point Clouds
https://arxiv.org/abs/1812.05784
2020.Pillar-based Object Detection for Autonomous Driving
https://arxiv.org/abs/2007.10323
2020.AFDet: Anchor Free One Stage 3D Object Detection
https://arxiv.org/abs/2006.12671

回答下面这些问题：
1. Lidar based Object Detection 的几种代表方案是什么？
2. Lidar based Object Detection 和 Radar based Object Detection 区别在哪里？类似的是啥？
3. 你有什么理解不了的问题
4. 如果让你提出 Radar based Object Detection 的方案，你觉得有什么好的 idea？有什么可以从 Lidar based Object Detection 参考借鉴？

e). (Lidar) Tracking in BEV/3D Space
阅读以下论文：
2020.Probabilistic 3D Multi-Object Tracking for Autonomous Driving
https://arxiv.org/abs/2001.05673
https://github.com/eddyhkchiu/mahalanobis_3d_multi_object_tracking
2020.Probabilistic 3D Multi-Modal, Multi-Object Tracking for Autonomous Driving
https://arxiv.org/abs/2012.13755
2020.Center-based 3D Object Detection and Tracking
https://arxiv.org/pdf/2006.11275.pdf
代码：https://github.com/tianweiy/CenterPoint

回答下面这些问题：
1. 这几篇论文都是做什么的
2. 区别在哪里
3. 主要的方案是什么
4. 你有什么理解不了的问题

5. 如果让你实现这几篇论文的代码，你觉得还有哪些具体实际的问题不理解需要帮助

2020.A two-stage data association approach for 3D Multi-object Tracking
https://arxiv.org/abs/2101.08684
这篇论文总结了一些方案，看了之后可以和我们讲一下这些内容：
1. Table 1 中提到了很多内容，总结一下 Table 1 具体讲了什么？
2. Tracking 的性能受什么影响？
3. mAP 和 AMOTA 是什么？怎么定义？

f). 综述阅读
阅读以下综述论文：
2020.MMW Radar-Based Technologies in Autonomous Driving: A Review
本文主要是 Radar 在自动驾驶领域的应用方案的综述，包括了 grid map based processing(for object detection and localization), object detection, tracking, localization。

看了之后可以和我们讲一下这些内容：
1. 这篇综述论文具体讲了哪些内容，分别讲了哪些技术方案
2. 这篇综述论文读完之后给你的感觉作为综述不足之处在哪里
3. 你认为我们的综述论文和本文的区别在哪

2021.Review on Vehicle Detection Technology for Unmanned Ground Vehicles
本文主要是综述各种传感器在(低速)自动驾驶领域 object detection 的综述。

看了之后可以和我们讲一下这些内容：
1. 这篇综述论文具体讲了哪些内容，分别讲了哪些技术方案
2. 你对自动驾驶中的 object detection 有了哪些更系统的认识
3. 这篇综述论文读完之后给你有哪些启发

Tasks until 20210531
Get familiar with the open source data and PyTorch. Write the baseline PyTorch code to apply YOLOv4(anchor based) and CenterNet(anchor free as well, use center point of object. See paper: 2019.Objects as Points) on radar BEV data(discrete pseudo image) and evaluate the performance. In details, it include following steps:
0. First, decide which dataset we should go for:
0.1. Check nuScenes dataset(which contains radar detection points, and 3D/2D bounding box in BEV as ground truth label), get understanding the details of nuScenes data format. **If choose this dataset**, study how to write Python/PyTorch code to serve as interface for loading nuScenes dataset into suitable format, e.g. image sync with radar detection points, discrete radar detection points on bird eye's view into pseudo image.
0.2. Check CARRADA dataset(which contains radar range azimuth heat map and range Doppler map, and 2D bounding box in BEV for both range-azimuth heatmap and range-Doppler map as ground truth label), get understanding the details of CARRADA data format. **If choose this**

dataset, study how to write Python/PyTorch code to serve as interface for loading CARRADA dataset into suitable format, e.g. image sync with radar detection points, discrete radar detection points on bird eye's view into pseudo image.
See paper: 2020.CARRADA Dataset: Camera and Automotive Radar with Range-Angle-Doppler Annotations
0.3. Check RADIATE dataset(which contains radar range azimuth representation without Doppler information, and 2D bounding box in BEV as ground truth label), get understanding the details of RADIATE data format. **If choose this dataset**, study how to write Python/PyTorch code to serve as interface for loading RADIATE dataset into suitable format, e.g. image sync with radar detection points, discrete radar detection points on bird eye's view into pseudo image.
See paper: 2020.RADIATE: A Radar Dataset for Automotive Perception
0.4. Check CRUW dataset(which contains radar range azimuth representation with Doppler information, and point object classification label in BEV as ground truth label), get understanding the details of CRUW data format. **If choose this dataset**, study how to write Python/PyTorch code to serve as interface for loading CRUW dataset into suitable format, e.g. image sync with radar detection points, discrete radar detection points on bird eye's view into pseudo image.
See paper: 2020.RODNet: Radar Object Detection using Cross-Modal Supervision
0.5. Check RadarScenes dataset(which contains radar detection points, and point-wise classification label/segmentation as ground truth label), get understanding the details of CRUW data format. **If choose this dataset**, study how to write Python/PyTorch code to serve as interface for loading CRUW dataset into suitable format, e.g. image sync with radar detection points, discrete radar detection points on bird eye's view into pseudo image.
See paper: 2020.RadarScenes: A Real-World Radar Point Cloud Data Set for Automotive Applications

1. Secondly, understand radar based object detection task.
1.1. Read all reference papers(**See table I&II in paper: 2021.Automotive Radar Based Perception in ADAS and Autonomous Driving: A Review**), focus on details of “how they did it”.
1.2. Conclude all the approaches described in these papers, also conclude other papers if possible, think about what are the advantages have been introduced? What are the drawback still left there? Then check all the ideas proposed above and discuss each of these ideas.

2. Then get familiar with the open source data and PyTorch. Write the baseline PyTorch code to apply anchor based(e.g. modification based on YOLOv4) and anchor free(e.g. modification based on FCOSv2, and modification based on CenterNet) on radar BEV data(discrete pseudo image) and evaluate the performance. In details, it include following steps
2.1. Read the paper and other possible materials regarding YOLO v4(and you may need to read technical intro to YOLOv1 to v3) and think about how to apply/modify it for BEV radar data. Here are the useful materials:
2019.YOLOv4: Optimal Speed and Accuracy of Object Detection
https://zhuanlan.zhihu.com/p/183261974
https://mp.weixin.qq.com/s/vzb6qPqVnNHn0zqGkpj0kg
https://www.bilibili.com/video/BV1Q54y1D7vj
一个可能的从 YOLOv4 方案针对 cruw 数据集的 object detection 修改为：vovnet backbone +

channel attention + yolov4 neck + anchor circle based detection head.

怎么求取 anchor shape 与 gaussian heat map 之间的相似性？现在想到的俩种方案：

a). 用单峰的 GMM 模型去拟合每一个 ground truth 对应的 confidence map，同时 anchor shape 也用 anchor Gaussian distribution 实现。这样预测出来的结果就是一个以中心点为均值，方差 sigma 的 Gaussian distribution。这时 localization loss 可以考虑不用 IOU 而用 Mahalanobis distance 求预测出来的 Gaussian distribution 和 ground truth 用单峰 GMM 模型拟合出来的 confidence map 的"距离"作为一个主要 loss 项。

甚至，不需要设置 anchor shape，直接用网络预测 confidence map 的每个元素的值。这样一来就变成了用 YOLOv4 的 backbone 和 neck 的 anchor box free object detector 了。

在这种方案中我们的 loss 只需要由 2 部分组成：

1. confidence map loss：这部分可以完全不区分正负样本，而是像原文那样（包括 CenterNet 也是这样处理）软化正负样本的区别（即用 Gaussian 分布的 confidence map 来替代 hard decision 的正：1，负：0 的概念），换言之就是直接对 ground truth 的 confidence map 和网络估计的 condifence map 之间求 FOCAL loss 的 BCE loss 即可。然后 classification loss 这一部分可以集成到 confidence map loss 内，也是不需要 hard decision 的方式区别正负样本，而是像 confidence map loss 一样用 Gaussian 分布的这种软化的概念。换言之，我们的网络甚至不需要在输出的 feature map 的每个 pixel 上预测 size 为（4 + 1, number_of_class）的预测结果。比如在 feature map 的 size 和输入 input BEV heat map 数据 size 相同的情况下，只需让每个 pixel 输出 size 为（1, number_of_class）的预测结果即可，这里的 1 对应的就是那个 pixel 的 confidence score 即可。也就是说，classification loss 其实已经集成到 confidence map loss 里面了。这个其实就是 CenterNet 中的 heatmap loss 项。

2. localization loss：这一部分也是仅对某些"正样本"成立，但因为这里我们没有定义什么是正样本什么是负样本，所以可以做这样一个设置：先对预测出来的 confidence map 求取 N 个值最大的 index，然后把这 N 个最大值的 index 和周边的一定区域的预测出来的 confidence map 的元素值用 GMM 拟合一个高斯分布，分别和 Ground truth condifence map 中 GMM 拟合出来的高斯分布求 mahalanobis distance，选和每一个 ground truth cofidence map 中对用的每一个高斯分布 mahalanobis distance 最小的作为"正样本"，然后用对应的"正样本"的 mahalanobis distance 作为 localization loss。

b). 用一个 bounding box 去框住 ground truth 对应的 confidence map，这样就又把 ground truth 变为了 bounding box 的形式表征了。如此一来，还是用 YOLOv4 原始的 anchor box 的方案设置，GIOU 作为 localization loss 项即可。

2.2. Read the paper and other possible materials regarding FCOSv2(and v1) and think about what is the difference between YOLO framework and FCOS? What is the difference between generic anchor based and anchor free object detector? What should be modified in detection head for CRUW dataset（Beware CRUW dataset does NOT have bounding box but only point ground truth）? how to apply/modify it for BEV radar data? Here are the useful materials:

2019.FCOS: Fully Convolutional One-Stage Object Detection (v1)

https://mp.weixin.qq.com/s/DOjHFvA1UfHWGHFcS7xjLw

https://mp.weixin.qq.com/s/o48-aYI_C5offve5uMjuNA

2020.FCOS: A Simple and Strong Anchor-free Object Detector (v2)

https://mp.weixin.qq.com/s/n-XQRxP6lpj4LK-1z5Xepw

2.3. Read the paper and other possible materials regarding CenterNet(Using Center Point of Object) and think about what is the difference between FCOS framework and CenterNet? What

should be modified in detection head for CRUW dataset（Beware CRUW dataset does NOT have bounding box but only point ground truth, **should CenterNet be the perfect matching for CRUW dataset? Maybe, especially check the detail of how to apply the CenterNet on estimation of key points of human pose, this application has lots of similarity compared with radar based point object detection using CRUW dataset**）? how to apply/modify it for BEV radar data? Here are the useful materials:

2019.Objects as Points

https://mp.weixin.qq.com/s/c2LJGuNG4wLuVzFG7gDhMg

https://mp.weixin.qq.com/s/wWqdjsJ6U86lML0rSohz4A

一个可能的将 CenterNet 应用在 CRUW 数据集上的修改为：去掉 CenterNet 原来的 detection head 的 3 个输出分支中的 height and width regression 分支，仅保留 Heat Map prediction 分支与 Offset 分支(Offset 分支是为了因为 stride = 4 的图像缩放导致的精度损失而设计的)。总结下来就是：vovnet backbone + channel attention + centernet neck + heatmap based center point only detection head

2.4. Write the code and write the detailed comments to describe the functionality for the code.

2.6. Adjust anchor based and anchor free deep learning object detector a bit to make it performs well for different data formats. Compare the results and discuss the advantages and drawbacks of different data formats.

2.7. Consider other improvements and ideas proposed in **Possible Ideas and Tasks to Do**, or propose new ideas for better novelty.

2.8. Write the paper of this work with focusing on discussion of what influence is brought to the object detection result by using different data formats.

The Following Plan(Possibly) as Step 2(Radar based Perception: Object Detection and Tracking in BEV/3D Space):

1. Extend the radar based object detector to radar based perception(i.e. object detection and tracking in BEV/3D space).

1.1. Basically two possible frameworks:

a). Object detection first, then tracking by using detections provided by radar based object detector.

b). End to end tracking, radar data in, track information output. Thus the end to end approach integrates the object detection and tracking in BEV/3D space together.

1.2. Write the code and do the experiment.

1.3. Consider other improvements and ideas proposed in **Possible Ideas and Tasks to Do**, or propose new ideas for better novelty.

1.4. Write the paper of this work with focusing on discussion of what influence is brought to the radar perception result.

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